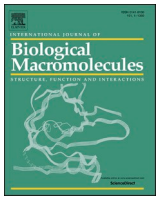
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Synergistic release of lignin-carbohydrate structure by Lewis acid cations and anions to enhance delignification in deep eutectic solvents

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ABSTRACT

Lewis acids are commonly used in deep eutectic solvents (DESs) to enhance delignification in lignocellulose pretreatment, though the mechanism is not well understood. This study compares the effects of alkali metal chlorides, acidic metal chlorides, and acidic metal non-chlorides on DESs delignification, focusing on metal ions and DESs properties. Results indicate that delignification enhancement follows the order: acidic metal chlorides > acidic metal non-chlorides > alkali metal chlorides. Key factors include pH (Pearson correlation coefficient: −0.88) and hydrogen bond donor activity (correlation: 0.91), and other DESs properties, such as viscosity, surface tension, hydrogen bond acceptor activity, and polarity, provide limited explanatory for delignification performance. The anion (Cl^-) forms hydrogen bonds with lignin and lignin-carbohydrate complexes (LCCs) hydroxyl groups and further participates in ether bond cleavage. The destruction of β-O-4 linkages in recovered lignin confirms this finding, alongside the disruption of LCCs. Cations primarily facilitate the hydrolysis of hemicellulose glycosidic bonds by directly cracking and lowering the bond energy (−3.02 kJ/mol). This leads to the release and further degradation of LCCs, especially in γ-ester, *p*-coumaric acid, and ferulic acid components. These findings provide insight into Lewis acid-enhanced DESs delignification mechanisms.

1. Introduction

Due to the non-renewable nature of fossil energy sources and their contribution to the greenhouse effect, there is a growing urgency to develop sustainable and clean alternatives. Lignocellulosic biomass is considered a promising candidate to replace fossil fuels due to its abundance and high calorific value. However, the complex physical structure of cellulose, hemicellulose, and lignin in lignocellulosic biomass limits its accessibility [1–3]. A "lignin-first" strategy is critical for the efficient utilization of lignocellulose [4,5]. Various pretreatment methods have been developed, among which deep eutectic solvents (DESs) are favored for their efficiency, recyclability, low corrosivity, and environmental friendliness [6]. Lewis acids, particularly metal chlorides, are often used as additives to enhance the delignification effect of DESs [7,8]. Additionally, Lewis acids can serve as catalysts for furfural production [9]. The mechanism by which Lewis acids enhance the delignification capacity of DESs has not been thoroughly investigated, nor has the structural transformation of the three major components of lignocellulose after DES pretreatment been systematically studied. In conclusion, understanding the pretreatment mechanism is crucial for maximizing process efficiency [6].

DESs are composed of hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs), with hydrogen bonding as the primary driving force. Their eutectic points exhibit significant negative deviations. It is widely believed that the delignification capacity of DESs results from the formation of competitive hydrogen bonds between the HBAs and HBDs in DESs and lignin. Lewis acids are electron-pair acceptors that readily bind to electron-rich molecules and anions [10], and any metal cation

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capable of accepting electrons is considered a Lewis acid. Lewis acids, especially strongly acidic metal chlorides, can be used as additives in choline chloride-based DESs to form ternary DESs or as the HBAs component in binary DESs with HBDs, such as carboxylic acids or polyols. Our previous study demonstrated that the Lewis acid enhances the pretreatment of lactic acid-based DESs, including delignification and furfural production. Existing literature also suggests that most Lewis acids, such as $CuCl_2$, $ZnCl_2$, and $FeCl_3$, enhance the delignification of DESs, with the mechanism attributed to the increased acidity and enhanced hydrogen bonding due to the Cl^- anion, as well as to covalent bond cleavage [8,11,12]. However, an interesting phenomenon is observed: the same molar concentration of different Lewis acids (e.g., $AlCl_3$ and $FeCl_3$) enhances the delignification capacity of DESs to differing extents, as shown in a literature survey [8,11]. If the enhanced delignification capacity were solely attributed to the action of Cl^-, no difference in delignification capacity would be observed between DESs containing the same molar concentration of anions. Currently, there is a lack of mechanistic exploration regarding the differences in delignification of DESs caused by Lewis acids, and the contributions of the physicochemical properties of Lewis acid anions, cations, and DESs to delignification remain unexplored.

Studies on DESs delignification mechanisms focus mainly on the weak interactions between Cl^- and lignin, with limited attention to the role of cations. Lewis acid cations are often identified as catalysts for xylose isomerization to produce furfural, restricting their perceived involvement in delignification. Tang et al. [8] examined $FeCl_3$-enhanced lactic acid-based DESs and observed that both hemicellulose removal and lignin dissolution follow a Boltzmann curve. Several studies [13–15] have highlighted that the removal of hemicellulose enhances lignin removal. Notably, in lignocellulosic biomass, hemicellulose is considered a bridge between lignin and cellulose components in the cell wall [16]. It interacts with cellulose through van der Waals forces and a limited number of hydrogen bonds, without forming covalent bonds. In contrast, lignin primarily forms lignin-carbohydrate complexes (LCCs) with hemicellulose. Consequently, lignin release mainly occurs through the disruption of LCCs structures or the cleavage of hemicellulose glycosidic bonds [5,11]. Xu et al. [17] reported a synergistic effect of metal cations on the selective separation of hemicellulose. Metal cations, on one hand, complex with water molecules to release hydrogen protons, and on the other hand, form complexes with carbohydrates, weakening intramolecular chemical bonds [18]. Based on these findings, we hypothesize that the cations in Lewis acids enhance the hydrolysis of hemicellulose, thereby strengthening the delignification capacity of DESs.

In this study, we investigated the effects of different metal salts on the delignification of DESs and systematically analyzed the physicochemical properties of DESs (e.g., pH, viscosity, surface tension, Kamlet-Taft parameters) along with the impact of Lewis acid cations and anions on delignification. The goal was to clarify the mechanism underlying the enhanced delignification ability of DESs facilitated by Lewis acids, especially the cation. Additionally, we explored the kinetic process of hemicellulose → furfural under the influence of Lewis acids, as well as the structural characteristics of the separated products (cellulose-rich materials and recovery lignin), providing a deeper understanding of the pretreatment process in Lewis acid-enhanced DESs.

2. Materials and method

2.1. Materials

Sugarcane bagasse (SCB) was obtained from the Institute of Biological and Medical Engineering, Guangdong Academy of Sciences, and sieved through a 100-mesh screen. Lactic acid (LA, CAS: 50-21-5), choline chloride (ChCl, CAS: 67-48-1), glycol (GO, CAS: 107-21-1), ferric chloride (FeCl3, CAS: 7705-08-0), aluminum sulfate ($Al_2(SO_4)_3$, CAS: 10043-01-3), ferric sulfate ($Fe_2(SO_4)_3$, CAS: 10028-22-5), copper chloride ($CuCl_2$, CAS: 7447-39-4), ferrous chloride ($FeCl_2$, CAS: 7758-94-3), magnesium chloride ($MgCl_2$, CAS: 7786-30-3), Methyl isobutyl ketone (MIBK, CAS: 108-10-1) and sodium chloride (NaCl, CAS: 7647-14-5) were provided by Sigma-Aldrich (Shanghai, China).

2.2. Preparation of DESs

The DES system used in this study (ChCl: LA: GO with a molar ratio of 1:5:1) was established due to the exceptional delignification efficiency of lactic acid-based DESs across various types of biomass, as demonstrated in numerous studies [19,20]. The ChCl, LA and GO were stirred in a water bath at 80 °C for 30 min until a clear, homogeneous liquid was formed. Metal salts were then added, maintaining a cation molar concentration of 0.185 mol/L, and the mixture was stirred at 70 °C until the salts were completely dissolved. Several metal salts, including $AlCl_3$, $FeCl_3$, $CuCl_2$, $FeCl_2$, $MgCl_2$, NaCl, $Fe_2(SO_4)_3$, and $Al_2(SO_4)_3$, were selected to clarify the influence of different Lewis acids on the delignification of DESs [8,11]. The samples were labeled as DESs/X, where X denotes the type of metal salt. pH modulation of the DESs was carried out using a saturated sodium hydroxide solution to minimize the impact of excessive aqueous solution on the properties of the DESs.

2.3. Pretreatment of SCB with DESs

The DESs were mixed with sugarcane bagasse in a 20:1 ratio and heated with stirring at 120 °C for 3 h. The mixture was then rinsed with three times its volume of distilled water at 80 °C. The wash solution was collected, and the pH was adjusted to 2 using 2 M hydrochloric acid. The solution was equilibrated at 25 °C for 24 h to allow complete precipitation of the lignin. The precipitated lignin was then centrifuged and rinsed three times with an ethanol solution (1:9, *v*/v) before being freeze-dried to obtain lignin powder. The cellulose-rich material was dried in an oven at 55 °C for 24 h and then weighed to calculate the solid recovery.

2.4. Recovery of DESs

Collect the solution after the precipitation of lignin from the pretreated bagasse and evaporate the water under rotation at 60 °C. Then, add MIBK reagent at a ratio of 1:2 to form a two-phase system and continue purifying the regenerated DESs for 30 min. After recovering the DESs, dry them in an oven at 40 °C and adjust the pH to be consistent with that of fresh DESs.

2.5. Chemical composition analysis

The determination of the chemical components of lignocellulose was based on a method reported by the National Energy Laboratory [21], which involved a two-step hydrolysis using H_2SO_4. The qualitative and quantitative analyses of monosaccharides, furfural, and furfuryl aldehydes were performed using a Waters HPLC system (Waters Corporation, Milford, MA, USA) with an Aminex HPX-87H column (7.8 mm × 300 mm, USA) [22].

Solids recovery, cellulose recovery, hemicellulose removal, delignification, and furfural yield were calculated using the following equations:

$$Cellulose\ recovery = \frac{W_{glycan\ in\ solid\ residue}}{W_{glycan\ in\ SCB}} \times 100\% \quad (1)$$

$$Hemicellulose\ removal = \frac{W_{xylan\ in\ SCB} - W_{xylan\ in\ solid\ residue}}{W_{xylan\ in\ SCB}} \times 100\% \quad (2)$$

$$Delignification = \frac{W_{lignin\ in\ SCB} - W_{lignin\ in\ solid\ residue}}{W_{lignin\ in\ SCB}} \times 100\% \quad (3)$$

$$Furfural\ yield = \frac{W_{furfural\ in\ pretreatment\ liquid}}{W_{xylose\ in\ SCB}} \times \frac{150}{96} \times 100\% \quad (4)$$

The quantitative determination of xylo-oligosaccharides (XOS) followed the method of Tang et al. [8]. Specifically, the H_2SO_4 concentration in the DES pretreatment solution was adjusted to 4.0 %, and the reaction was conducted at 121 °C for 1.0 h to fully hydrolyze the xylo-oligosaccharides. The amount of XOS was determined by comparing the xylan content before and after hydrolysis.

2.6. Characterization of physicochemical properties of DESs

The viscosity of the DESs was determined using a rheometer (RST-SST, Brookfield-SST, Middleboro, MA, USA). The samples were equilibrated for 5 min prior to testing, and the viscosity was measured at a shear rate of 0.1–100 s^{-1} at 25 °C. Surface tension was measured using a Dataphysics DCAT21 high-sensitivity microelectromechanical balance system. To quantify the hydrogen bond donors, hydrogen bond acceptors, and polarity of the DESs, Kamlet-Taft parameters were calculated. Specifically, Nile Red, 4-nitroaniline, and N,N-diethyl-4-nitroaniline were prepared in a 0.001 mol/L solution. A 100 μL aliquot of the dye solution was mixed with 4 mL of DESs solution, following the evaporation of methanol at 40 °C. The absorbance was then scanned from 350 to 700 nm using a UV spectrophotometer, and the maximum absorption wavelength was used to calculate α (hydrogen-bond acidity), β (hydrogen-bond basicity), and π* (polarity), as described by Tang et al. [8].

2.7. Quantum chemical computation

The molecular structures of lignin and hemicellulose with metal chlorides ($AlCl_3$ and $FeCl_3$) were drawn using ChemDraw software (PerkinElmer Informatics, Boston, MA, USA) and pre-optimized for energy minimization in the MM2 force field. The lignin model was constructed using the SGG model (Supplementary Material, Fig. S1A), and the hemicellulose model was constructed using the trimeric xylan model (Supplementary Material, Fig. S1B). Molecular docking sites and weak interaction analyses were performed following the methodology described by [3]. The lowest energy conformations of lignin/xylan with Lewis acids were identified using molecular dynamics in the xTB program [23], and these conformations were further optimized with the Gaussian 16 W software package (Gaussian, Inc., Wallingford, CT, USA) at the B3LYP-D3(BJ)/6-31G* level. The output "wfn" files were then analyzed for molecular weak interactions and visualized using the Multiwfn 3.8 program [24] based on the independent gradient model of Hirshfeld partition (IGMH).

2.8. Kinetic rate fitting for hemicellulose hydrolysis

The differential equations were solved by parameter fitting using the fmincon function in Matlab software (MATLAB R2024a). Record the removal amounts of xylose, xylan, furfural, and hemicellulose at 20-min intervals within 0 to 240 min, plot the kinetic curves, and fit the data to the kinetic model for parameter estimation.

2.9. Two-dimensional nuclear magnetic characterization of lignin structure

A 50 mg lignin sample was mixed with 0.5 mL of deuterated dimethyl sulfoxide (DMSO-d_6) with 0.03 % tetramethylsilane (TMS) and analyzed using a 600 MHz NMR spectrometer to acquire 2D HSQC spectra. The ^1H spectrum was sampled from 0 to 10 ppm with a relaxation delay of 1 s and 2048 sampling points, while the ^{13}C spectrum was sampled from 0 to 210 ppm with 256 sampling points and a scan count of 64. The ^1H—^{13}C chemical shifts were calibrated using the C—H cross-peak signal of the DMSO solvent at $\delta = 39.5/2.49$ ppm, and the spectra were processed using MestReNova software (Mestrelab Research S.L.).

2.10. Characterization of solid residue

To investigate the functional group and bond linkage differences in lignocellulosic structures before and after DES pretreatment, Fourier Transform Infrared Spectroscopy (FT-IR, Thermo Fisher Nicolet IS50, USA) was used to record response signals in the range of 4000 to 400 cm^{-1}. Changes in the amorphous regions of cellulose before and after DES treatment were analyzed by assessing the crystallinity of cellulose in the solid residue. X-ray diffraction (XRD) analysis was performed using a PANalytical Empyrean II diffractometer, with a scanning range of 5° ~ 50°. The crystallinity index (CrI) of the solid residue was calculated according to the method of Jiang et al. [25]. Surface morphology of the solid residue was examined via environmental scanning electron microscopy (SEM, Merlin, ZEISS).

3. Results and discussion

3.1. Effect of Lewis acids on the delignification capacity of DESs

To investigate the effect of different Lewis acids on the DESs pretreatment process, three types of Lewis acids were selected: acidic metal chlorides ($AlCl_3$, $FeCl_3$, $CuCl_2$), basic metal chlorides ($FeCl_2$, $MgCl_2$, NaCl), and acidic metal non-chlorides ($Fe_2(SO_4)_3$, $Al_2(SO_4)_3$), which were individually added to lactic acid-based DESs. As shown in Fig. 1A, the cellulose content of the cellulose-rich materials after DESs treatment increased due to the effective separation of hemicellulose and lignin. Furthermore, the addition of metal salts further enhanced the cellulose content in the cellulose-rich materials after pretreatment. Interestingly, while the molar amount of metal cations added was consistent across all metal salts, the DESs/$AlCl_3$ group exhibited the highest cellulose content at 64.00 %, along with a delignification rate of 93.15 %, indicating the most effective pretreatment compared to other metal salt systems (Fig. 1A). This conclusion is consistent with previous studies [8,11].

DESs/NaCl exhibited a delignification capacity of 77.2 %, slightly higher than DESs without metal salt (75.8 %), but no significant difference was observed (Fig. 1B). The addition of other metal salts to the DESs system resulted in significantly enhanced delignification compared to DESs alone. The ranking of metal salts in terms of enhancing the delignification capacity of DESs is as follows: acidic metal chlorides > acidic metal non-chlorides > alkali metal chlorides. Previous studies have attributed the enhanced delignification capability of DESs to acidic metal chlorides, particularly the role of Cl^- [5,8], which is also confirmed in this study. For example, among ferric salt with different anions, DESs/$FeCl_3$ (91.1 %) showed a higher delignification capacity than DESs/$FeCl_2$ (84.77 %) and DESs/$Fe_2(SO_4)_3$ (85.36 %). Similarly, DESs/$AlCl_3$ (93.15 %) demonstrated a higher delignification capacity than DESs/$Al_2(SO_4)_3$ (87.68 %), highlighting the importance of Cl^- in enhancing the delignification capacity of DESs. However, within the acidic metal chlorides, a significant difference in delignification enhancement was observed between DESs/$FeCl_3$ and DESs/$AlCl_3$ at the same molar concentration, as well as between DESs/$FeCl_2$, DESs/$CuCl_2$, and DESs/$MgCl_2$. Even metal salts without Cl^- (e.g. $Fe_2(SO_4)_3$, $Al_2(SO_4)_3$) also promoted the delignification capacity of DESs, suggesting that while Cl^- is a key factor influencing delignification, it is not the sole factor. Similarly, in the cetyltrimethylammonium bromide (CTAB)-based DESs [8] and glycerol-based DESs [11], Cl^- is not considered the sole factor determining the delignification differences among DESs.

To elucidate the impact of cation differences on the delignification capacity of DESs, the pKa values of metal ions were collected from the work of Kamireddy et al. [26] and are presented in Fig. 1C. The trend of metal cation pKa values mirrors the trend observed in their ability to enhance the delignification capacity of DESs. For example, with a consistent concentration of the anion (Cl^-), the pKa values for Cu^{2+}, Fe^{2+}, and Mg^{2+} are 6.50, 9.49, and 11.4, respectively, with a

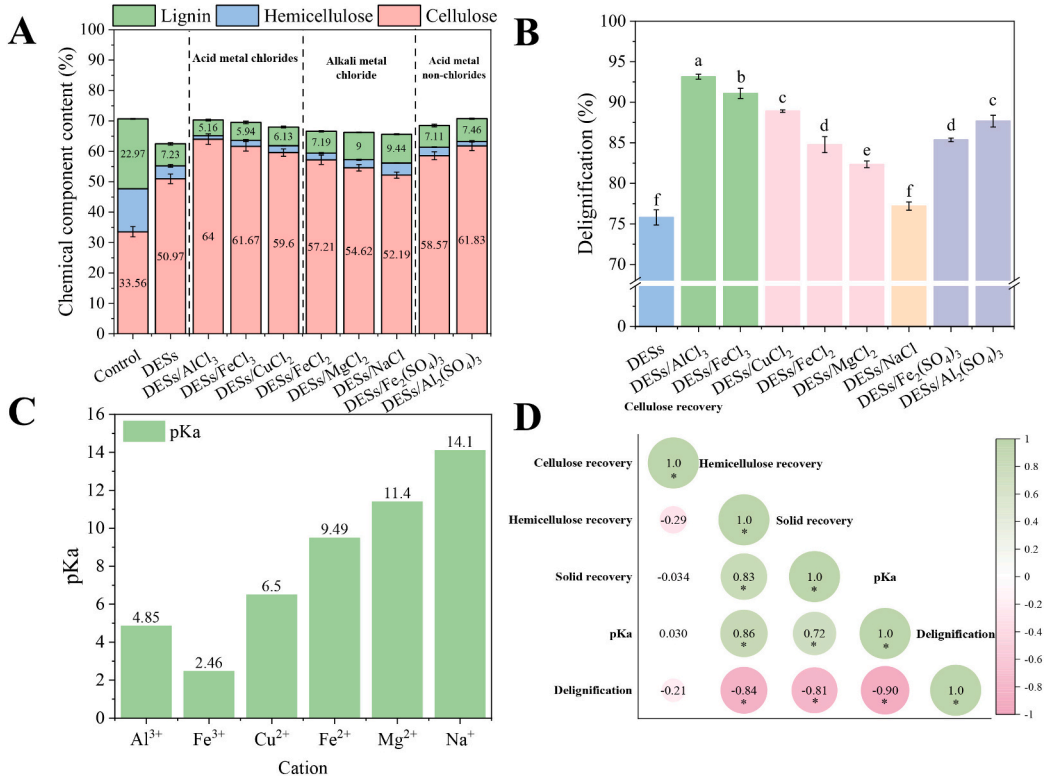


Fig. 1. Effect of Lewis acid metal salts (acidic metal chlorides, alkali metal chlorides, and acidic metal non-chlorides) with different pKa value on the pretreatment of DESs and the correlation analysis. (A) Compositional analysis of cellulose-rich fractions; (B) Comparative delignification analysis. Statistical significance was assessed by one-way ANOVA, with different letters indicating significant differences (Duncan test, $p < 0.05$). (C) pKa values of metal cations; (D) Correlation analysis, where the values presented are correlation coefficients (* indicates significance at $p < 0.05$). In the numbering, the chemical formula after each slash represents the type of metal salt added to the DESs.

corresponding gradual decrease in delignification capacity. However, this trend does not hold for Al^{3+} and Fe^{3+}, where Fe^{3+} has a lower pKa value (2.46) compared to Al^{3+} (4.85), yet Al^{3+} exhibits the strongest delignification effect. Therefore, while pKa values influence the varying enhancement effects of metal chloride salts on the delignification capacity of DESs, they do not fully explain why DESs/$AlCl_3$ (93.15 % delignification) shows stronger delignification capability than DESs/$FeCl_3$ (91.1 % delignification), which is consistent with the conclusion reported by Tang et al. [8].

Fig. 1D illustrates the correlation between cellulose recovery, hemicellulose recovery, solid recovery, the pKa of cations, and delignification. The results indicate the highest Pearson correlation coefficient between pKa and delignification, reaching −0.90 ($p < 0.05$, Fig. 1D). Notably, a smaller pKa value corresponds to a stronger ability of the metal salt to dissociate hydrogen protons, which indirectly leads to changes in the physicochemical properties of DESs (e.g., a decrease in pH). Hemicellulose recovery and solid recovery also exhibit significant correlations with the delignification capacity of DESs, with correlation coefficients of −0.84 and − 0.81, respectively. The significant correlation between solid recovery and delignification is attributed to the effective removal of impurities from the lignocellulosic material during DESs pretreatment, which reflects the intensity of the pretreatment. The observed correlation between hemicellulose recovery and delignification is consistent with findings from several studies [13–15,27], which have shown that the removal of hemicellulose enhances lignin removal. This effect is attributed to the disruption of the cross-linked lignocellulose structure (e.g., LCCs). Tang et al. [8] found that both delignification and hemicellulose removal rates follow an "S" shaped Boltzmann curve when discussing the effects of different severity factors. Similar trends have been observed in the studies by Huo et al. [11] and He et al. [5]. In summary, besides the influence of the metal salt anion (Cl^-), changes in the physicochemical properties of DESs induced by the pKa value of cations, as well as the inherent relationship between hemicellulose and lignin removal, may also contribute to the delignification capacity of DESs. This requires further investigation.

3.2. *Mechanism discussion of the influence of DESs physicochemical properties on delignification*

To facilitate comparison, five DESs systems were selected to further investigate the changes in lignin removal induced by the alterations in the physicochemical properties of DESs due to the presence of Lewis acids. In this study, the pH, viscosity, surface tension, and hydrogen bonding interactions of DESs were considered. Viscosity influences the mass transfer efficiency of DESs, thereby affecting the contact efficiency between the solvent and solute [20]. Surface tension affects the permeability of DESs [8], while hydrogen bonding is a driving force for lignin removal in DESs [28].

The pH of DES systems with $FeCl_3$ and $Fe_2(SO_4)_3$ is lower than that of DESs with $AlCl_3$ and $Al_2(SO_4)_3$, which is consistent with the trend observed for pKa values (Supplement material Table S1). DESs/$FeCl_3$ has the lowest pH (−0.44), but its delignification ability is lower than that of DESs/$AlCl_3$ (pH = −0.24). Moreover, DESs/$AlCl_3$ and DESs/$Fe_2(SO_4)_3$ exhibit similar pH values, yet their lignin removal capabilities differ, which is consistent with the findings of Huo et al. [11]. Based on these phenomena, speculation is made that metal cations also contribute to the delignification process of DESs, with the discussion in subsequent research.

Supplementary material Table S1 shows the changes in the physicochemical properties of DESs induced by Lewis acids. Generally, higher viscosity tends to hinder the lignin removal effectiveness of DESs. The addition of $AlCl_3$, $Al_2(SO_4)_3$, and $Fe_2(SO_4)_3$ increased the viscosity of DESs

by 29.73 mPa·s, 8.13 mPa·s, and 4.92 mPa·s, respectively. Obviously, the increase in viscosity of DESs caused by $AlCl_3$ is more pronounced. Therefore, viscosity cannot explain the enhanced lignin removal ability of DESs. Moreover, $FeCl_3$ and $Fe_2(SO_4)_3$ have a stronger effect on increasing the surface tension of DESs than $AlCl_3$ and $Al_2(SO_4)_3$, with increases of 0.36 and 0.77 mN/m, respectively. Thus, the influence of metal salts on the surface tension of DESs offers limited explanation for the improved lignin removal ability. The Kamlet-Taft (K-T) parameters, which quantify the hydrogen bond donor (acidity, α), hydrogen bond acceptor (basicity, β), and polarity (π*), show correlations between α [29] and α-β [30] with lignin solubility, thereby partially explaining the lignin removal ability [31]. The α values of DESs/$AlCl_3$ and DESs/$FeCl_3$ were 2.61 and 2.63, respectively, while the β values were 1.56 and 1.31, which are higher than that of DESs (α = 1.68, β = 1.27), indicating that the addition of chloride salts facilitates the formation of a richer hydrogen bond network within DESs. Notably, $Al_2(SO_4)_3$ and $Fe_2(SO_4)_3$ effectively enhance the hydrogen bond donor capacity of DESs but have a limited effect on the hydrogen bond acceptor capacity. This is likely because the electrostatic interactions between the metal cations and DESs molecules alter the polarity of the DESs, thereby affecting the hydrogen bond donor ability. Since Cl^- is the primary hydrogen bond acceptor in DESs, metal salts containing Cl^- enhance the hydrogen bond acceptor ability (β value) of DESs. Fig. 2 shows the correlation analysis between the physicochemical properties of DESs and their lignin removal ability, revealing a significant negative correlation between pH and lignin removal (correlation coefficient = −0.88) and a significant positive correlation between α (hydrogen bond donor) and lignin removal (correlation coefficient = 0.91). In fact, both pH and hydrogen bond donor ability (α) are indicators of the proton donation ability of DEScoc, so it is reasonable that both pH and α values are significantly correlated with lignin removal in DESs.

Overall, changes in the physicochemical properties of DESs, especially the proton donation ability, are the primary factors influencing lignin removal, while the effects on viscosity, surface tension, hydrogen bond acceptor capacity, and polarity are insufficient to cause significant changes in delignification ability. The proton donor ability can explain the stronger delignification enhancement of acidic metal salts compared to basic metal salts, but it cannot account for the superior delignification promotion by aluminum salts over iron salts.

3.3. Effect of anions of metal salts on the delignification capacity of DESs

As expected, adjusting the pH of all DESs systems to 0.60 significantly reduced their delignification capacity (Fig. 3A), confirming the critical role of pH in influencing DESs-mediated delignification. Under the same pH conditions, the delignification capacities of DESs/$AlCl_3$, DESs/$Al_2(SO_4)_3$, DESs/$FeCl_3$, and DESs/$Fe_2(SO_4)_3$ were 71.59 %, 65.57 %, 66.71 %, and 60.7 %, respectively. Both Lewis acid cations and anions contribute to the delignification of DESs. The delignification differences resulting from anionic variations were 6.01 % and 6.02 %, which were greater than those caused by cations (4.87 % and 4.88 %). For anions (Cl^-), previous studies [5,32–34] indicate that the α-hydroxy group of lignin initially undergoes protonation, and Cl^- nucleophilically attacks the α-hydroxy group following protonation, forming a chloride intermediate and further facilitating ether bond cleavage in lignin [32,35](Supplementary Fig. S2A). In this work, quantum chemical calculations of weak interactions between metal salts and lignin (Supplementary Fig. S2B) reveal that Cl^- forms hydrogen bonds with the α- and γ-hydroxy groups of lignin. This suggests that Cl^- has the potential to compete with lignin in forming hydrogen bonds, thereby disrupting the hydrogen-bond network of lignocellulose, as reported by Xia et al. [28]. No weak interactions were observed between cations and the lignin bonding, suggesting that anions play a more significant role in the structural breakdown of lignin compared to cations. The weak interactions between LCCs and Lewis acids are illustrated in Supplementary Material Fig. S3. The results demonstrate that Cl^- readily forms hydrogen bonds with hydroxyl groups in the LCCs linkage bonds; however, no evidence was observed for weak interactions between cations and the oxygen atoms in the linkage bonds of LCCs, either through weak interaction analysis or in previously reported studies. Notably, the bond energy of β-O-4 linkages is only −1.20 kJ/mol, which is lower than that of LCCs, indicating that lignin depolymerization occurs preferentially before the disruption of LCCs. Overall, Cl^- can promote lignin cleavage through the formation of hydrogen bonds with lignin, and nucleophilic attack on the protonated α-site hydroxyl group.

3.4. Mechanism of Lewis acid cation-promoted delignification

As shown in Fig. 1D, the recovery of hemicellulose is significantly negatively correlated with delignification. The hemicellulose removal rate for DESs/$AlCl_3$ (pH control) is 91.0 %, which is 10.5 % and 3.4 %



Fig. 2. Correlation analysis of the effect of metal salts on the physicochemical properties of DESs related to delignification. The values shown represent correlation coefficients, with * indicating significance at $p < 0.05$.

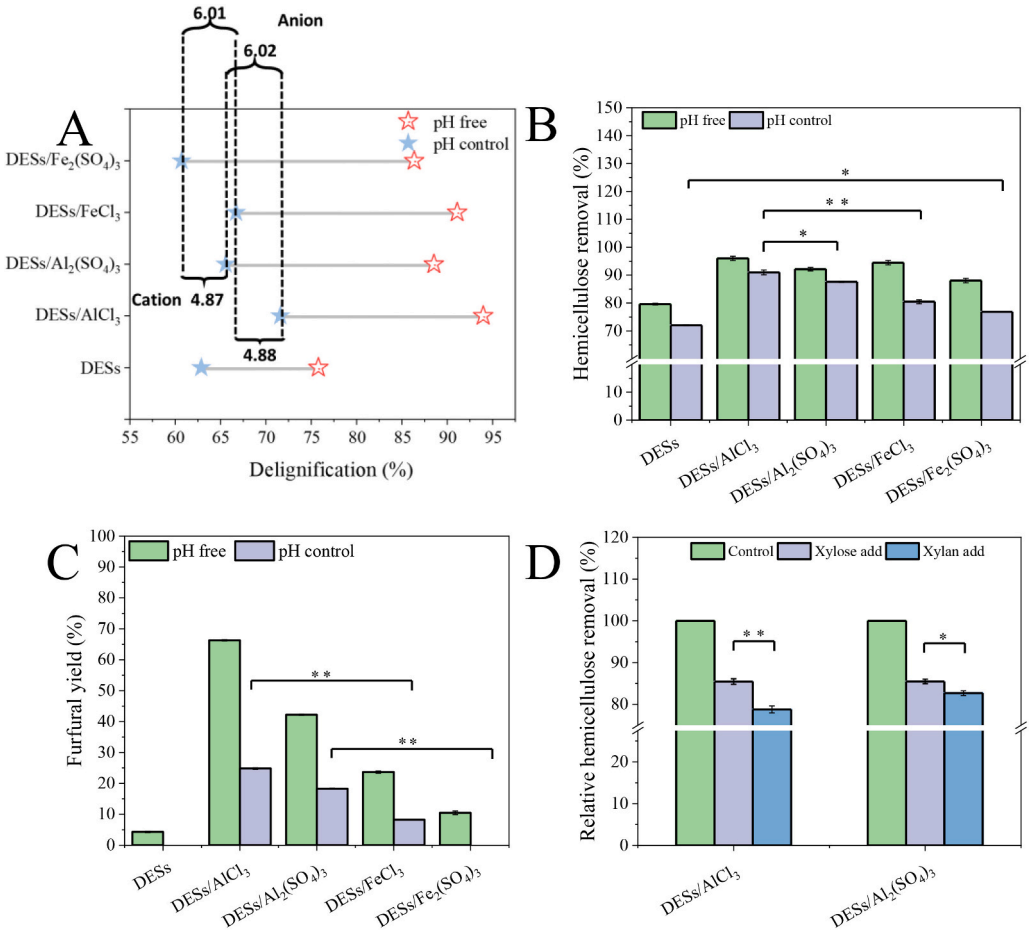


Fig. 3. Mechanism by which metal salts enhance the delignification capacity of DESs. (A) Contribution of metal salt anions and cations to the delignification capacity of DESs; (B) Hemicellulose removal rate; (C) Furfural conversion rate; (D) Effect of the additional of 0.07 mol/L xylose and xylan on the hemicellulose remove capacity of DESs. "pH free" refers to the mixing of Lewis acid with DESs without pH adjustment, while "pH control" refers to the adjustment of pH to 0.60 ± 0.01. * represents significance at $p < 0.05$, ** represents significance at $p < 0.01$.

higher than those for DESs/$FeCl_3$ (pH control) and DESs/$Al_2(SO_4)_3$ (pH control), respectively. Therefore, the effect of cations on hemicellulose removal is greater than that of anions (Fig. 3B).

To elucidate the possible mechanism by which cations enhance hemicellulose removal, the independent gradient model based on Hirshfeld partition (IGMH) was applied to assess the non-covalent interactions between Lewis acids and xylan. Supplemental Figs. S4 illustrate the scatter plot and isosurface plot of the IGMH, where the blue region represents the attractive interactions between Lewis acids and xylan, the green region corresponds to van der Waals forces, and the red region represents repulsive interactions. Notably, the cations of Lewis acids (Fe^{3+}, Al^{3+}) preferentially interact weakly with the oxygen on the xylan pyranose ring, suggesting that metal ions primarily promote the pyranose ring opening reaction, which is a prerequisite for furfural production. And in this study, it was observed that DESs containing aluminum salts generated more furfural and had higher hemicellulose removal rates compared to those containing iron salts (Fig. 3C). Although weak interaction analysis did not directly observe the effect of cations on the glycosidic bond, some studies suggest that the attraction between metal ions and the oxygen atom of the glycosidic bond can lead to chemical bond stretching, which facilitates glycosidic bond hydrolysis [36–38]. Through bond energy calculations, ring-opening reduces the bond energy of glycosidic bonds by 0.4 kJ/mol (Supplement material Table S2). Therefore, we propose that cations promote the pyranose ring opening reaction, which, in synergy with their effect on the glycosidic bond, facilitates hemicellulose hydrolysis.

Al^{3+} exhibits superior ability to promote hemicellulose hydrolysis compared to Fe^{3+} (Fig. 2B). This difference in cation performance can be explained by the ionization energy of metal chlorides, as demonstrated by Lee et al. [39]. The performance of metal ions is evaluated using the χ/r^2 parameter, where χ represents electronegativity and r denotes ionic radius. The χ/r^2 value for Al^{3+} is 5.23, while for Fe^{3+} it is 4.13, indicating that Al^{3+} has a stronger Lewis acidity, which facilitates the dehydration and isomerization of xylose. In this work, the IGMH analysis results confirm that the attractive interaction (hydrogen bonding) between Al^{3+} and xylan is even more pronounced than that of Fe^{3+}. And the distances between Al^{3+} and Fe^{3+} and the oxygen atoms on the xylose pyranose ring are 1.90 Å and 2.10 Å, respectively (Supplementary Fig. S4C), further confirming that Al^{3+} exhibits stronger catalytic capability in the opening reaction of the pyranose ring.

To verify the mechanism by which cations enhance hemicellulose hydrolysis, we added exogenous 0.07 mol/L xylose to competitively react with the cations, thereby inhibiting their effect on the pyranose ring opening of xylose/xylan derived from lignocellulose. The results shown in Fig. 3D indicate that the relative hemicellulose removal after the addition of xylose to DESs/$AlCl_3$ and DESs/$Al_2(SO_4)_3$ decreased by 14.55 % and 14.52 %, respectively, while the relative delignification rates also decreased by 8.16 % and 8.42 % (Supplementary Fig. S5). These differences are statistically significant, demonstrating the effect of cations on the pyranose ring opening of hemicellulose, which further impacts delignification in DESs. Furthermore, the addition of xylan at the same molar concentration reduced the relative hemicellulose

removal rates of DESs/$AlCl_3$ and DESs/$Al_2(SO_4)_3$ by 21.2 % and 17.31 %, respectively (Fig. 3D), and significantly impaired the lignin removal efficiency. If cations only affected the pyranose ring opening reaction, the impact of adding xylan and xylose on hemicellulose removal by DESs should be consistent. However, the experimental results show that the inhibition of lignin removal by exogenous xylan is more pronounced than that by xylose, indicating that cations also contribute to the direct hydrolysis of glycosidic bonds.

By comparing the bond dissociation energies of glycosidic bonds and LCCs (Supplementary Material, Table S2), it is observed that the bond energy of glycosidic bonds (−3.42 kJ/mol) is higher than that of lignin's β-O-4 structure (−1.20 kJ/mol) and LCCs linkages (phenyl glycoside: −3.03 kJ/mol; γ-ester: −2.99 kJ/mol; ferulic acid: −3.00 kJ/mol). The bond energy of glycosidic linkages in hemicellulose is higher than that of LCCs and lignin ether bonds. This may explain why previous studies have primarily focused on the depolymerization of LCCs and lignin rather than the hydrolysis of hemicellulose.

3.5. Kinetics of hemicellulose hydrolysis

To further understand the hydrolysis process of hemicellulose during pretreatment with DESs by pH and cation/anion of metal salts, time-product concentration curves of hemicellulose in DESs were plotted (Supplementary Fig. S6). In order to more effectively describe the hydrolysis of hemicellulose in DESs, the product evolution pathway of hemicellulose in DESs was mapped based on the work of reported literature [40], which indicated the possible reaction pathway that were involved in the reaction system (Scheme 1).Where k_1 is the rate constant for the separation of hemicellulose from the lignocellulosic structure to form xylan; theoretically, during the pretreatment process, hemicellulose is not ideally completely separated from the lignocellulose and then hydrolyzed, and xylan here is considered a hypothetical intermediate. The k_2, k_3, and k_4 are the rates of hydrolysis of xylan to form monosaccharides and XOS, and of hydrolysis of XOS to xylose, respectively. Ershova et al. [18] confirmed a two-pathway model for the conversion process of xylose to furfural. Here, k_5, k_6 represent the rate constants for isomerization of xylose to intermediates and further degradation to furfural, respectively. k_7 represents the direct dehydration of xylose to produce furfural. K_8, k_9, k_{10} and k_{11} represent the rate constants for the formation of humin from xylose, intermediates, furfural, and xylose + furfural, respectively; k_{12} is the rate constant for the separation of LCCs from lignocellulose. Based on this model, nine differential equations were developed to describe the transformation of hemicellulose:

$$\frac{\partial x_1}{\partial t} = -k_1 x_1 - k_{12} x_9 \tag{5}$$

$$\frac{\partial x_2}{\partial t} = k_1 x_1 - (k_2 + k_3) x_2 \tag{6}$$

$$\frac{\partial x_3}{\partial t} = k_4 x_4 + k_3 x_2 - (k_5 + k_7 + k_8) x_3 - k_{11} x_3 x_6 \tag{7}$$

$$\frac{\partial x_4}{\partial t} = k_2 x_2 - k_4 x_4 \tag{8}$$

$$\frac{\partial x_5}{\partial t} = k_5 x_3 - (k_6 + k_9) x_5 \tag{9}$$

$$\frac{\partial x_6}{\partial t} = k_6 x_5 + k_7 x_3 - k_{10} x_6 - k_{11} x_3 x_6 \tag{10}$$

$$\frac{\partial x_7}{\partial t} = k_8 x_3 + k_9 x_5 + k_{10} x_6 \tag{11}$$

$$\frac{\partial x_8}{\partial t} = k_{11} x_3 x_6 \tag{12}$$

$$\frac{\partial x_9}{\partial t} = k_{12} x_1 \tag{13}$$

Through parameter fitting, Supplementary Table S3 summarizes the fitting results of all the rate constants. In carboxylic acid-based DESs, the rate of hemicellulose removal, producing xylan, reaches k_1 = 5.32, which is faster than the subsequent hydrolysis of xylan. This indicates that hemicellulose is more easily removed in lactic acid-based DESs, demonstrating high selectivity. Additionally, the reaction rates for xylan → oligoxylan, xylan → monosaccharides, and oligoxylan → monosaccharides (k_2 = 4.11, k_3 = 3.18, k_4 = 3.38) are much higher than those for humic substance formation (k_8 = 2.13, k_9 = 0.25, k_{10} = 0.40), which is consistent with the findings of Yang et al. [40].

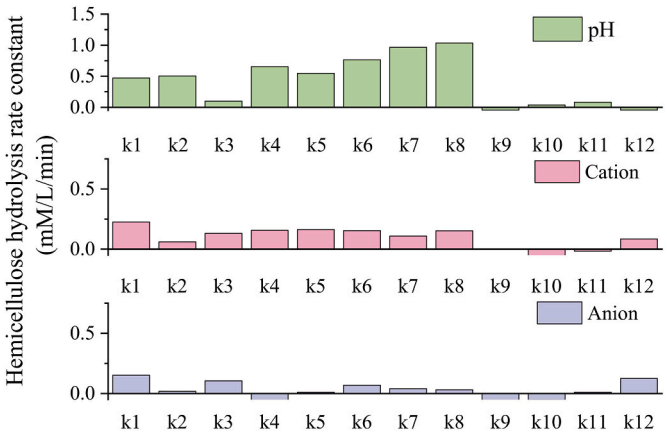
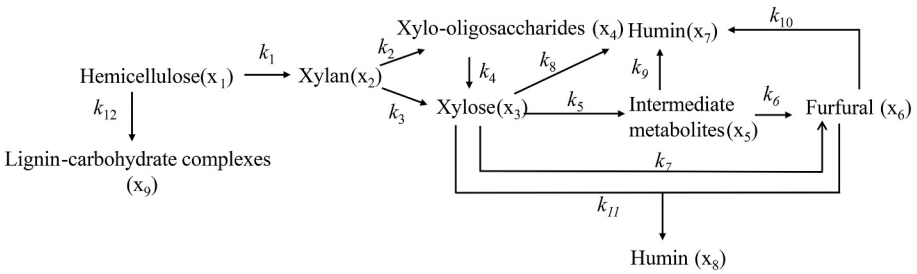


Fig. 4. Effect of pH, cation, and anion on the hemicellulose hydrolysis kinetic parameters. The effect of pH on the kinetic parameters is calculated by subtracting the kinetic parameters of DESs/$AlCl_3$ (pH free) from those of DESs/$AlCl_3$ (pH control). The effect of cation is determined by subtracting the kinetic parameters of DESs/$AlCl_3$ (pH control) from those of DESs/$FeCl_3$ (pH control). The effect of anion is calculated by subtracting the kinetic parameters of DESs/$AlCl_3$ (pH control) from those of DESs/$Al_2(SO_4)_3$ (pH control). A higher value indicates a greater impact of pH, cation, or anion on the respective process.



Scheme 1. Reaction model used to determine the kinetic rate constants of hemicellulose conversion in carboxylic acid DESs in the presence of Lewis acid (here, hemicellulose is represented as xylose only; other sugars are not considered). k represents the kinetic rate constant.

Fig. 4 illustrates the effects of pH, cations, and anions on the hemicellulose hydrolysis kinetics. The higher the difference in k, the more significant the influence of pH value, cation or anion on the reaction [41,42]. For example, the pH has a greater impact on the hydrolysis of hemicellulose and the formation of furfural compared to the generation of humin, due to the larger differences in the values of $k_1 \sim k_2$ and $k_4 \sim k_7$ than those of $k_9 \sim k_{12}$. The pH promotes the hemicellulose hydrolysis rate, including hemicellulose removal (k_1) and depolymerization (k_2, k_3, k_4), due to the proton attack on the hemicellulose glycosidic bonds. pH also facilitates the xylan → intermediate metabolites transition (k_5, from 3.06 to 3.60), indicating that low pH and Al^{3+} synergistically catalyze the xylan ring opening. No effect of pH on k_{12} was observed, which may be due to the hydrolysis of LCCs under acidic conditions, thereby reducing the formation of LCCs. From the perspective of cations, Al^{3+} increases k_3 and k_4 by 0.13 and 0.16, respectively, compared to Fe^{3+}, which is higher than the increase in k_2 (0.06). In other words, cations can promote the conversion of polysaccharides into monosaccharides and oligosaccharides, and the *exo*-acting rate of polysaccharides is greater than the *endo*-acting rate. This confirms the direct pathway through which cations promote hemicellulose hydrolysis. It is noteworthy that the differences in Lewis acid cations also affect k_{12}, which confirms that the hydrolysis of hemicellulose glycosidic bonds promotes the release of LCCs, thereby facilitating the removal of lignin. Anion-mediated effects were also observed, with chloride ions (Cl^-) promoting the release of the LCCs structure ($k_{12} = 0.12$). This suggests that the competitive hydrogen bonding interaction between Cl^- and lignin aids in the disruption of the lignocellulosic structure, leading to further LCCs release, which in turn promotes the release of hemicellulose ($k_1 = 0.15$).

3.6. Altered chemical structure of recovery lignin under the influence of Lewis acid

Previous studies have confirmed that lactate-based ionic liquids have the ability to decompose LCC (including benzyl esters, benzyl ethers, and β-O-4) in lignocellulose. [43]. In this work, two-dimensional nuclear magnetic resonance (2D NMR) was used to characterize the structure of lignin recovered from Lewis acid enhanced-DESs (Fig. 5). As reported by Ji et al. [44], lignin from sugarcane bagasse mainly contains methoxyl groups (OMe), aryl ether linkages (A, β-O-4), β-β resinols (B), and phenylcoumaran (C, β-5), with the aromatic rings mainly composed of guaiacyl (G), syringyl (S), p-coumaric acid (PCA), and ferulic acid (FA). The β-O-4 linkage content in DESs/$AlCl_3$ (pH control) was 35.29/100Ar, which is 31.01/100Ar higher than that in DESs/$AlCl_3$ (pH free), indicating that pH significantly disrupts the ether bonds in lignin. Interestingly, under consistent pH, the β-O-4 content in DESs/$AlCl_3$ and DESs/$FeCl_3$ containing Cl^- was 35.29/100Ar and 33.36/100Ar, respectively, showing no significant difference. In contrast, DESs/$Al_2(SO_4)_3$ without Cl^- exhibited the highest β-O-4 retention (46.73/100Ar), further confirming that Cl^- participates in the cleavage of ether bonds, while cations have a minimal effect on lignin ether bond degradation. The formation of condensed lignin is highly sensitive to pH, with an upregulation in pH reducing the condensed lignin content by 26.31/100Ar.

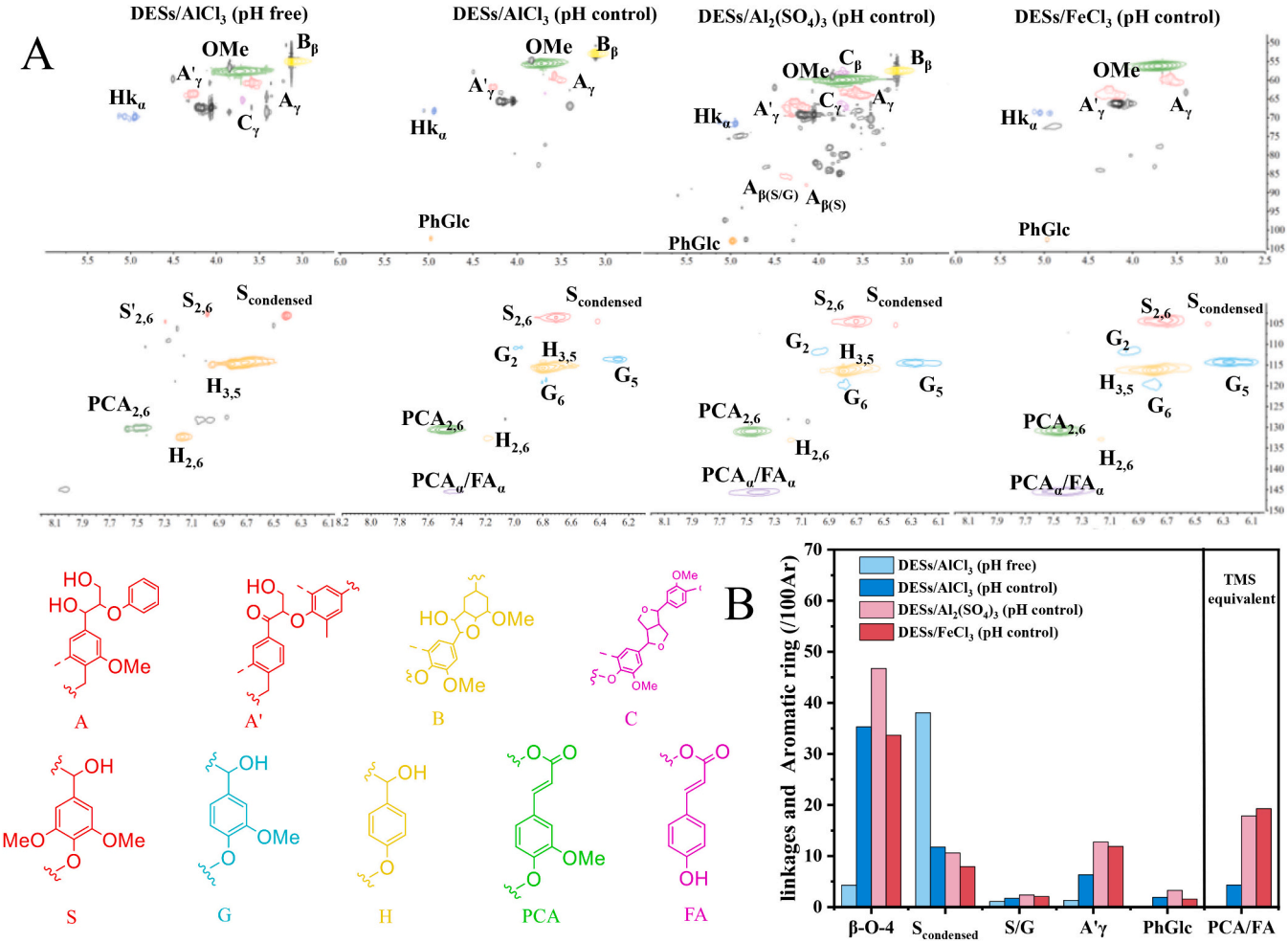


Fig. 5. The two-dimensional NMR analysis was used to investigate the lignin linkage and aromatic ring content in different DESs systems. (A): Signals from the side-chain and aromatic regions; (B): Summary of signals. PCA/FA signals were relatively quantified using 0.03 % tetramethylsilane (TMS) as an internal standard.

Additionally, cation variations have a greater effect on condensed lignin formation than anion variations. Both cations and anions significantly impact the G2, G5, and G6 signals. Analysis of the S/G ratio (Fig. 5B) reveals that the G unit undergoes preferential cleavage, with Cl^- having a stronger cleavage effect on the G unit than cations. Therefore, Cl^- is more effective than cations in cleaving lignin ether bonds and the G unit, but the subsequent lignin condensation process is more influenced by cations. This provides new insights into the mechanism behind Lewis acids enhancing the formation of condensed lignin.

PhGlc (phenyl glycoside) and A'γ (γ-ester) are considered to be the connection bonds between xylan and lignin [45,46]. Under low pH conditions, the contents of PhGlc and A'γ are 0/100Ar and 1.32/100Ar respectively, indicating that pH significantly enhances the degradation of LCCs. Cl^- affects both PhGlc and A'γ, whereas the differences in cations influence only A'γ (Fig. 5). This phenomenon may be attributed to the hydrogen bonding interactions between Cl^- and LCCs, which directly disrupt the LCCs structure. In contrast, cation-induced hemicellulose hydrolysis primarily facilitates the detachment of LCCs from lignocellulose, thereby increasing their exposure to DESs. Consequently, the impact on PhGlc, which has a higher bond energy (−3.03 kJ/mol), is relatively minor. Furthermore, PCA and FA are also typical LCCs bonds between lignin and xylan [25], TMS was used as an internal standard for the relative quantification of PCA/FA. FA and PCA units are covalently cross-linked with polysaccharides through ester bonds and with lignin components primarily through ether bonds. Both cations and anions affect FA and PCA, with the influence of cations on PCA/FA being slightly pronounced than that of anions, as observed through the PCAα/FAα signals. Overall, pH is the main factor driving lignin depolymerization and LCCs degradation. Additionally, Cl^- have a greater effect on lignin ether bond cleavage and the G unit, and Cl^- disrupts the LCCs structure (PhGlc, γ-ester, FA, and PCA). Cations primarily facilitate the hydrolysis of hemicellulose, further promoting the disruption of γ-ester and PCA/FA (LCCs), and exert a greater influence on the formation of condensed lignin.

3.7. Structural characterization of the cellulose-rich fraction under the influence of Lewis acid

The FTIR spectra of sugarcane bagasse after DES treatment are shown in Fig. 6A. The spectral peaks at 3300–3500 cm^{-1} correspond to the stretching vibrations of hydroxyl groups. Lignin aromatic ring skeletal vibrations are observed at 1600 cm^{-1} and 1510 cm^{-1}, while C—H vibrations linked to the benzene ring appear at 826 cm^{-1}. The lignin signals in cellulose-enriched material pretreated with DESs supplemented with $AlCl_3$ are significantly attenuated, indicating lignin removal. And, raising the pH resulted in an enhancement of lignin signal peaks in the cellulose-enriched material. The results of the chemical composition analysis also indicate that increasing the pH is detrimental to lignin removal (Fig. 6B). The signal peak at 1100 cm^{-1}, attributed to the C—O stretching vibrations of cellulose, becomes more pronounced in the cellulose-enriched material after DESs treatment. This suggests that the increased proportion of cellulose (Fig. 6B) results in greater exposure of its structural components. In comparison, cellulose signals in materials treated with DESs/$FeCl_3$ (pH control) and DESs/$Al_2(SO_4)_3$ (pH control) are weaker than those treated with DESs/$AlCl_3$ (pH control), indicating that both cations and anions in metal salts influence the depolymerization of lignocellulosic cross-linked structures. The lignin-hemicellulose signal appears at 1250 cm^{-1}. For the untreated raw material (Control), this lignin-hemicellulose signal is prominent. The use of DESs enhanced with Lewis acids effectively removes lignin-hemicellulose structures from lignocellulose. After treatment with DESs/$AlCl_3$ (pH free), this signal disappears, but it reappears when the pH is increased, indicating that pH plays a role in disrupting lignin-hemicellulose structures. Furthermore, it was observed that DESs/$FeCl_3$ (pH control) and DESs/$Al_2(SO_4)_3$ (pH control) caused a significant reduction in the lignin-hemicellulose signal compared to the raw material. This suggests that both anions and cations contribute to the disruption of lignin-hemicellulose structures, which aligns with the findings from hemicellulose hydrolysis kinetics.

The addition of Lewis acids increased the crystallinity of cellulose from 42.34 % to 57.59 % (Fig. 6C). Cations had a significant effect on cellulose crystallinity, with the crystallinity difference between DESs/$AlCl_3$ (pH control) and DESs/$FeCl_3$ (pH control) reaching 4.22 %. This change was notably greater than the differences caused by anions (0.2 %) and pH (0.9 %), indicating that cations have a more pronounced impact on amorphous cellulose. Notably, hemicellulose primarily connects with amorphous cellulose, and the disruption of amorphous cellulose by cations facilitates the separation of LCCs from lignocellulose. Similarly, Yuan et al. [7] observed that aluminum chloride-enhanced acidic DESs effectively disrupt amorphous cellulose regions, releasing nanocellulose crystals. SEM analysis revealed that the untreated sugarcane bagasse exhibited a smooth surface with groove-like structures. After DESs treatment, the surface of the cellulose-rich material displayed an increased number of pores (Supplementary Fig. S7). Notably, the pore formation was more pronounced in materials treated with DESs containing Al^{3+} compared to those treated with DESs containing Fe^{3+}.

3.8. Mechanism of Lewis acid-enhanced delignification in DESs

Based on the above results, a mechanism for the enhanced delignification of DESs by Lewis acids is proposed, as shown in Fig. 7. H^+ is fundamental to both hemicellulose hydrolysis and lignin depolymerization. It can participate in the cleavage of lignin ether bonds and hemicellulose hydrolysis, while also synergizing with Lewis acid cations

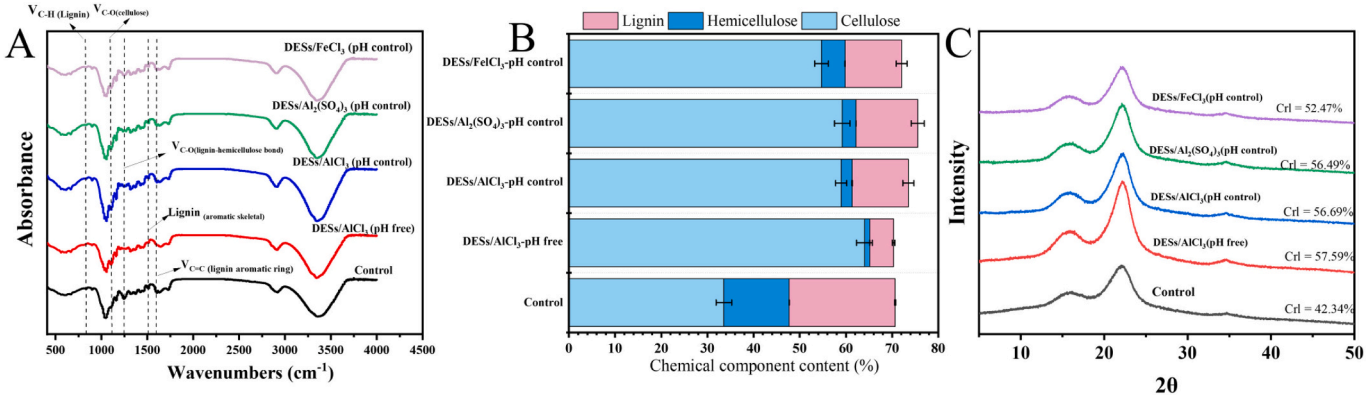


Fig. 6. Characterization of cellulose-rich fractions after Lewis acid pretreatment. A: Fourier-transform infrared (FTIR); B: Chemical component content; C: Crystallinity.

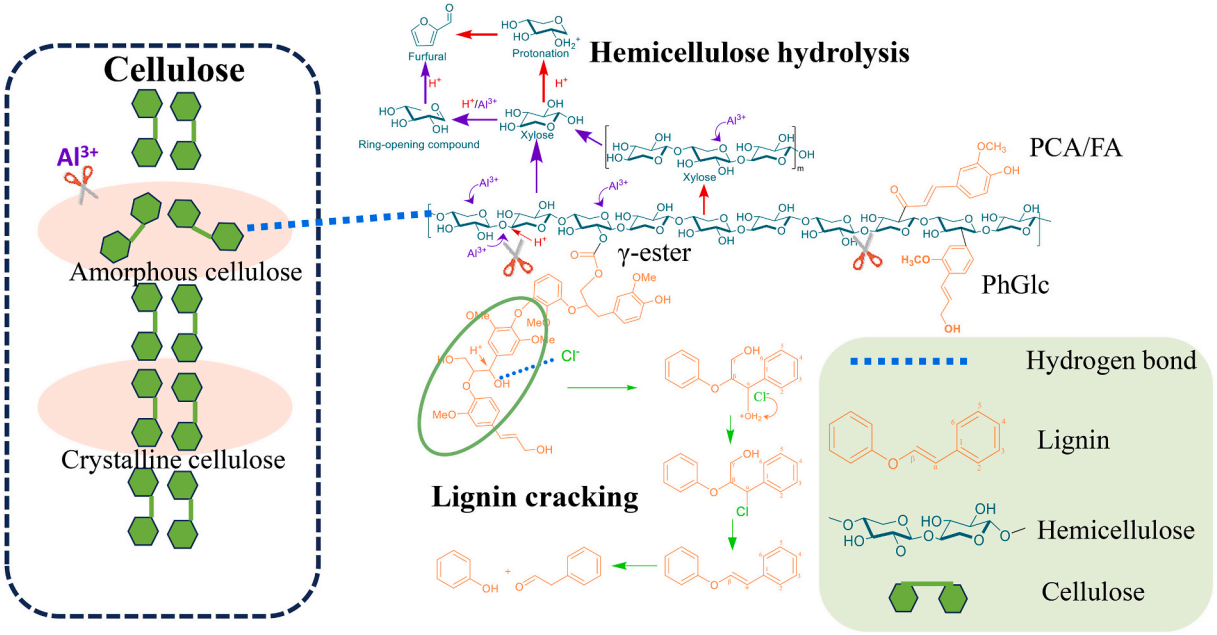


Fig. 7. Summary of the mechanism by which Lewis acids enhance the delignification capacity of DESs.

(Al^{3+}) and anions (Cl^-) to further enhance the disruption of LCCs, thereby promoting lignin removal. Initially, Cl^- competitively forms hydrogen bonds with the hydroxyl groups of lignin and LCCs, disrupting the hydrogen bond network within lignocellulose. Simultaneously, H^+ protonates the α-hydroxy group of lignin, enabling Cl^- anions to participate in the subsequent cleavage of lignin ether bonds. Cations primarily lower the bond energy of glycosidic linkages through the ring-opening of hemicellulose pyranose rings and directly contribute to glycosidic bond hydrolysis, accelerating the conversion of hemicellulose into furfural. In lignocellulose, lignin is predominantly connected to hemicellulose via LCCs structures. Therefore, the hydrolysis of hemicellulose glycosidic bonds also leads to the release of LCCs, and this conclusion was supported by studies on the kinetics of hemicellulose hydrolysis. These released LCCs structures are further degraded in DESs, particularly targeting γ-ester and PCA/FA structures, while the higher bond energy of PhGlc renders it less affected. Therefore, Lewis acids enhance the H^+ donor capacity of DESs, and together with cations and anions, they promote hemicellulose hydrolysis and lignin depolymerization, further disrupting LCCs and facilitating more efficient lignin removal.

The recyclability of DESs was also evaluated to emphasize the application value of the Lewis acid-enhanced DESs pretreatment. By enhancing the in-situ conversion of hemicellulose and purifying the regenerated DESs using the organic solvent methyl isobutyl ketone (MIBK), the relative delignification of the recycled DESs and furfural yield decreased by 3.42 % and 1.98 % over ten cycles, respectively (Supplement material Fig. S8). This excellent recyclability is attributed to not only the effective removal of impurities (e.g. Hemicellulose degradation products) but also the stability of the recycled DESs. Shen et al. [19] have confirmed through nuclear magnetic resonance carbon spectrum and two-dimensional nuclear magnetic resonance that within the pretreatment temperature range of 90–130 °C, the recovered lactic acid-based DES still maintains a stable chemical structure. The stable recyclable performance makes the Lewis acid-strengthened DESs have excellent application prospects.

4. Conclusion

This study demonstrates that the effect of different Lewis acids on the delignification ability of DESs is influenced not only by pH (proton concentration) and Cl^- anions but also by the ability of cations to accelerate hemicellulose hydrolysis, which in turn contributes to the lignin removal. Specifically, cations catalyze the cleavage of hemicellulose glycosidic bonds in conjunction with the ring-opening reaction of pyranose structures, effectively releasing the LCCs structure and thereby promoting lignin removal. Analysis of the recovered lignin structure indicates that cations do not affect the ether bonds in lignin but increase the condensation of lignin structures. Although no weak interactions between cations and LCCs were observed, it was found that the disruption of low-energy covalent bonds (e.g., γ-ester and PCA/FA) in the recovered lignin due to the exposure of LCCs in DESs. In cellulose-rich materials, cations exhibit a stronger ability to disrupt amorphous cellulose, suggesting that the weak interactions between cellulose and hemicellulose are broken, which accelerates the depolymerization of lignocellulose. Overall, the synergistic action of the anions and cations of Lewis acids enhances the disruption of LCCs structures, leading to more efficient lignin removal. This work provides new insights into the mechanism of Lewis acid-enhanced DESs for delignification.

CRediT authorship contribution statement

Qinghao Zhao: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Ming-Jun Zhu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Lu Zhao:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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://www.freepik.com/), under the appropriate Freepik licensing terms for editorial use.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.141732.

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